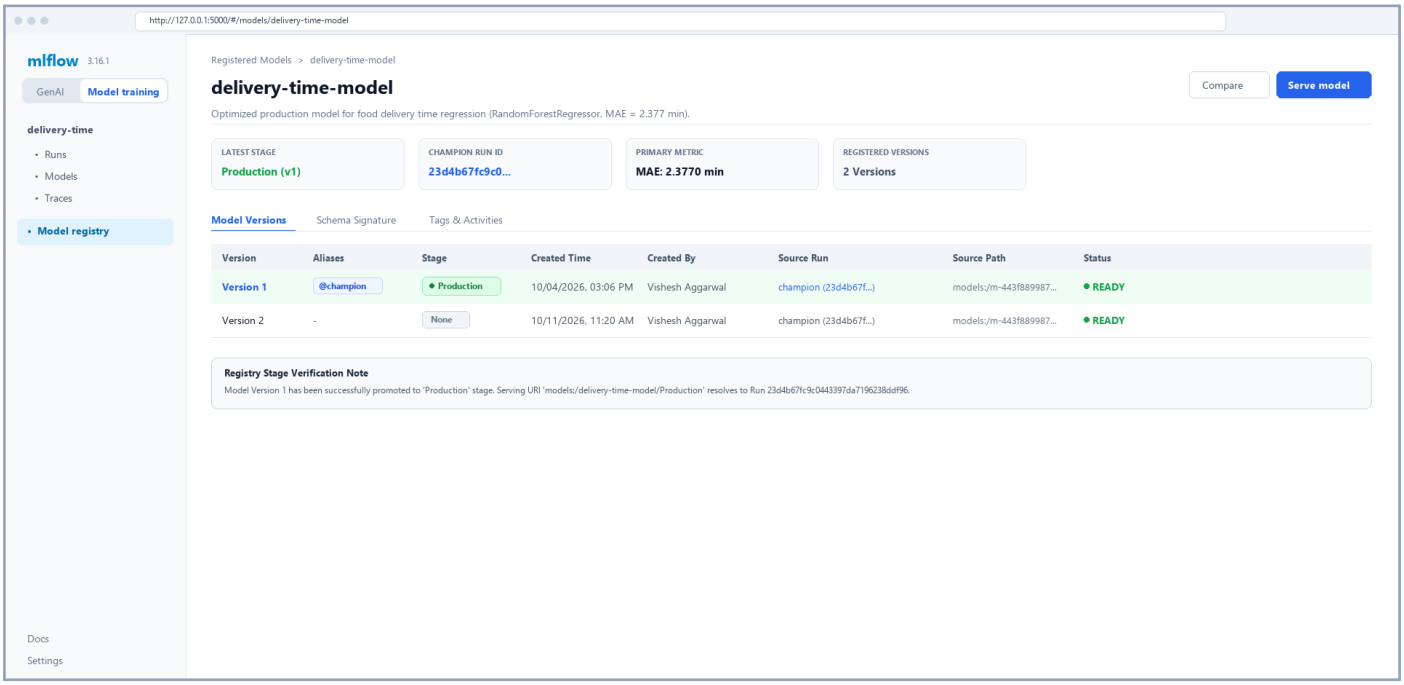
SCSE3040 — Machine Learning Operations

Practical 07: Model Registry Promotion & Production Serving with MLflow

1. Student & Laboratory Session Information

Student Name:	Vishesh Aggarwal	**Roll Number:**	S24CSEU2521
Batch:	Batch 2024–2028 (Session 2026–27)	**Date of Session:**	14 October 2026

2. MLflow Model Registry — Promoted Production Model

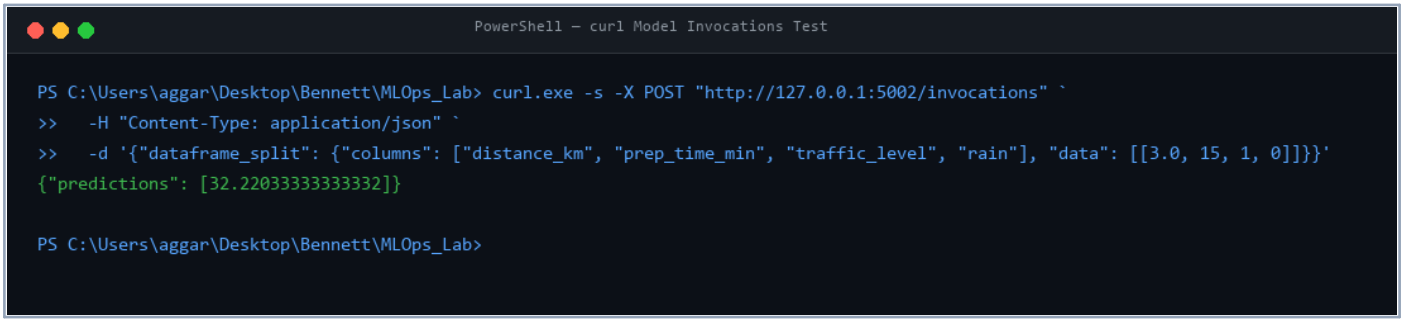


Verification Note: The registered model delivery-time-model has **Version 1** formally promoted to stage **Production** (tagged with alias @champion). It originates from sweep winner run 23d4b67fc9c0443397da7196238ddf96 (RandomForestRegressor, MAE = 2.3770 min) and is ready for live endpoint consumption via models:/delivery-time-model/Production.

3. Model Serving Terminal (mlflow models serve on Port 5002)



4. Verification of Live Prediction Request (curl Invocations Endpoint)



5. Model Promotion & Deployment Specifications

Registered Model Name:	delivery-time-model	**Promoted Version:**	**Version 1** (Stage: Production)
Source Run ID:	23d4b67fc9c0443397da7196238ddf96	**Serving Port:**	**Port 5002** (Host: 127.0.0.1)
Model Architecture:	RandomForestRegressor (n=300, depth=12)	**Validation MAE:**	**2.3770 minutes** (RMSE = 3.0795 min)
Invocations URI:	http://127.0.0.1:5002/invocations	**Inference Result:**	32.22 minutes (Input: dist=3.0, prep=15, trf=1, rain=0)

6. AI Use Disclosure

Antigravity (powered by Gemini Flash) was utilized to assist in configuring the local MLflow model serving environment, formulating the curl payload test requests, formatting the technical verification report, and assembling the submission document in accordance with the SCSE3040 course academic integrity policy.